Nilpotent Elements and the Jacobson–Morozov Theorem

$\mathfrak{g}$: complex finite dimensional Lie algebra

For each $x \in \mathfrak{g}$, define the adjoint representation

$ad : \mathfrak{g} \longrightarrow End(\mathfrak{g})$; $ad_x(\cdot) = [x, \cdot]$.

Def.: A complex finite-dimensional Lie algebra is called **reductive** if $ad : \mathfrak{g} \to End(\mathfrak{g})$ gives a completely reducible representation of $\mathfrak{g}$ on itself (i.e., $\mathfrak{g}$, viewed a $\mathfrak{g}$-module via ad, is a direct sum of irreducibles).

Obs.: 1) $\mathfrak{g}$ is reductive iff $\mathfrak{g} = z(\mathfrak{g}) \oplus [\mathfrak{g}, \mathfrak{g}]$ and $[\mathfrak{g}, \mathfrak{g}]$ a semisimple ideal of $\mathfrak{g}$.

2) Every semisimple Lie algebra is trivially reductive.

semisimple $\Rightarrow$ no nontrivial solvable ideals $\Rightarrow z(\mathfrak{g}) = 0$.

Def. (Killing form): Let $\mathfrak{g}$ be a Lie algebra. For each $X, Y \in \mathfrak{g}$, define the bilinear form $\kappa(X,Y) = Tr(ad_X\, ad_Y)$. Then κ is a symmetric bilinear form on $\mathfrak{g}$, called the **Killing form**. We have that κ is also invariant, in the sense that $\kappa([X,Y],Z) = \kappa(X,[Y,Z])$.

Fact: $\mathfrak{g}$ is semisimple $\Leftrightarrow$ κ is nondegenerate on $\mathfrak{g}$.

Def.: Let $\mathfrak{g}$ be a reductive Lie algebra.

a) We say an element $X \in \mathfrak{g}$ is **nilpotent** if $X \in [\mathfrak{g}, \mathfrak{g}]$ and ad_X is a nilpotent endomorphism on $\mathfrak{g}$. That is,

$(ad_X)^r = ad_X \circ ad_X \circ \cdots \circ ad_X = 0$ for some $r > 0$.

b) We say $X \in \mathfrak{g}$ is semisimple if $\operatorname{ad} X$ is semisimple (i.e., diagonalizable) as a linear endomorphism on $\mathfrak{g}$.

Let V be a finite dimensional vector space over $\mathbb{C}$ and denote by $\mathfrak{gl}(V)$ the Lie algebra of $\operatorname{End}(V)$ with $[X, Y] = XY - YX$.

The most important example of reductive Lie algebra is $\mathfrak{gl}_n = \mathfrak{gl}(\mathbb{C}^n)$. For semisimple Lie algebras, the most important example is $\mathfrak{sl}_n = \{X \in \mathfrak{gl}_n \mid \operatorname{trace}(X) = 0\}$.

Note that $\mathfrak{gl}_n = \mathbb{C} I_n \oplus \mathfrak{sl}_n$.

The following theorem is a very standard linear algebra fact that will be used throughout the talk.

Theorem 1 (Jordan decomposition).

Let V be a finite dimensional vector space over $\mathbb{C}$ and consider $X \in \operatorname{End}(V)$. Then

a) There exist unique operators $X_s, X_n \in \operatorname{End}(V)$ satisfying the conditions: $X = X_s + X_n$, where X_s is semisimple, X_n is nilpotent and $[X_s, X_n] = 0$

b) Each of the operators X_s and X_n are polynomials of X with zero constant term. In particular, X_s and X_n commute with any operator commuting with X and stabilize any subspace of V that X stabilizes.

Theorem 2: If X and X' in $\mathfrak{gl}(V)$ be such that $X' = PXP^{-1}$, $P \in GL(V)$, then $\operatorname{ad}_{X'} = \operatorname{Ad}_P \circ \operatorname{ad}_X \circ \operatorname{Ad}_{P^{-1}}$, where $\operatorname{Ad}_Q : \mathfrak{gl}(V) \to \mathfrak{gl}(V)$ denotes conjugation by $Q \in GL(V)$.

Proof: $ad_{X'}(Y) = [X', Y] = [PXP^{-1}, Y] = P[X, P^{-1}YP]P^{-1}$

$= Ad_P([X, P^{-1}YP]) = Ad_P(ad_X(P^{-1}YP))$

$= Ad_P \circ ad_X \circ Ad_{P^{-1}}(Y)$. □

Hence $ad_{X'}$ and ad_X are conjugate as elements of $gl(V)$ and thus they have the same spectrum, same Jordan form, same nilpotency/semisimplicity prop.

Theorem 3: If $X \in gl(V)$ is a nilpotent (resp., semisimple) endomorphism, then ad_X is also nilpotent (resp., semisimple).

Proof: a) Let $r > 0$ be such that $X^r = 0$. Associate with X the left and right multiplication operators (in $End(End V)$): $L_X(Y) = XY$ and $R_X(Y) = YX$. L_X and R_X are both nilpotent because X is; moreover, L_X and R_X commute. Now $ad_X = L_X - R_X$ is nilpotent because

$$(ad_X)^t = (L_X - R_X)^t = \sum_{k=0}^{t} (-1)^k \binom{t}{k} (L_X)^k (R_X)^{t-k},$$

which is zero for $t = 2r$.

b) Let X be semisimple and choose a basis $(v_1, \dots, v_n)$ of eigenvectors for V. Let $\{e_{ij}\}$ be the standard basis with respect to $\{v_i\}$ in $gl(V)$, i.e., $e_{ij} \cdot v_k = \delta_{jk} v_i$.

Then $X = \sum_{a=1}^{n} \lambda_a e_{aa}$ and

$$ad_X(e_{ij}) = [X, e_{ij}] = \left[\sum_a \lambda_a e_{aa}, e_{ij}\right]$$

$$= \sum_a \lambda_a [e_{aa}, e_{ij}] = (\lambda_i - \lambda_j) e_{ij},$$

and so ad_X can be written as a diagonal matrix in $gl(V)$. □

Corollary: Let $X \in gl(V)$ and let $X = X_s + X_n$ be its Jordan decomposition. Then $ad_X = ad_{X_s} + ad_{X_n}$ is the Jordan decomposition of ad_X (in $End(End(V))$).

Proof: We know ad_{X_s} is semisimple and ad_{X_n} is nilpotent, and they commute, because $[ad_{X_s}, ad_{X_n}] = ad_{[X_s, X_n]} = 0$. Therefore, the hypothesis of Theorem 1 applies. □

Note that since a semisimple Lie algebra g can be realized as a subalgebra of $End(g)$ via the adjoint representation, we can use the previous corollary to decompose each $X \in g$ as $X = X_s + X_n$, where $[X_n, X_s] = 0$. This is usually referred to as the Abstract Jordan Decomposition of X, and we define X_s to be the semisimple part and X_n the nilpotent part of X. It is a fact that both Jordan decompositions coincide.

Theorem 4: Given $X \in gl(V)$, ad_X is nilpotent if and only if $X = \lambda \cdot Id + A$, where A is a nilpotent operator.

Proof: Assume ad_X is nilpotent. Then the semisimple part of ad_X is zero: $ad_{X_s} = 0$. This means $[X_s, Y] = 0$ for all $Y \in gl(V)$. Take $\{v_1, \ldots, v_n\}$ a basis for V. Define the elementary operator E_{ij} such that $E_{ij}(v_k) = \delta_{jk} v_i$. Note that the E_{ij} span $gl(V)$. Write $X(v_j) = \sum_\ell x_{\ell k} v_\ell$.

Using $[X, E_{ij}] = 0$, we have $0 = X E_{ij}(v_k) - E_{ij} X(v_k)$.
If $k \neq j$, both sides are zero. If $k = j$, $0 = X(v_i) - E_{ij}\left(\sum_m x_{mj} v_m\right)$
$= \sum_\ell x_{\ell i} v_\ell - x_{jj} v_i$. For $\ell \neq i \to x_{\ell i} = 0$, so all zeros outside diagonal; for $\ell = i$, $x_{ii} - x_{jj} = 0$, so all elements in diagonal are equal. Therefore, $X = \lambda \mathrm{Id} + X_n$. Take $A = X_n$ and the proof follows.
Conversely, let $X = \lambda \mathrm{Id} + A$. Then
$\mathrm{ad}_X = \mathrm{ad}_{\lambda I} + \mathrm{ad}_A = \mathrm{ad}_A$, as $\lambda \mathrm{Id}$ is central. Since A is nilpotent, we have $\mathrm{ad}_A = \mathrm{ad}_X$ is nilpotent by Theorem 3. □

Corollary: $X \in \mathfrak{gl}_n$ is nilpotent iff X is a nilpotent (w.r.t. matrix multiplication) matrix.

Proof: Take $V = \mathbb{C}^n$ in Theorem 4. Then $X \in \mathfrak{gl}_n$ is nilpotent $X = \lambda I_n + A$, where A is a nilpotent matrix. If we need $X \in \mathfrak{sl}_n$, then we take $\lambda = 0$. The result follows. □

Def.: A standard triple in a semisimple Lie algebra $\mathfrak{g}$ is a triple of elements $X, Y, H \in \mathfrak{g}$ satisfying the relations

$$[H, X] = 2X, \quad [H, Y] = -2Y \text{ and } [X, Y] = H.$$

In other words, the subalgebra $\mathfrak{a} \subseteq \mathfrak{g}$ spanned by X, Y, H is isomorphic to $\mathfrak{sl}_2$. The elements H, X and Y are called respectively the semisimple (or neutral) nilpositive and nilnegative element of the triple $\{H, X, Y\}$.

Theorem 5: Let $\{X, Y, H\}$ be a standard triple in the semisimple Lie algebra $\mathfrak{g}$. Then X is a nilpotent element of $\mathfrak{g}$.

Proof: The commutators $[H, X] = 2X$, $[H, Y] = -2Y$ and $[X, Y] = H$ establish a Lie algebra homomorphism $\varphi : \mathfrak{sl}_2 \longrightarrow \mathfrak{g}$ such that $\varphi(e) = X$, $\varphi(f) = Y$ and $\varphi(h) = H$. Further, under the composition $\varphi \circ \mathrm{ad} : \mathfrak{sl}_2 \longrightarrow \mathrm{End}(\mathfrak{g})$ (ad is also a Lie algebra homomorphism), the algebra $\mathrm{End}(\mathfrak{g})$ is a finite-dimensional representation of $\mathfrak{sl}_2$.

As such, it is completely reducible:

$$\mathrm{End}(\mathfrak{g}) = \bigoplus_j V_j, \text{ where } V_j \text{ is an irrep of } \mathfrak{sl}_2.$$

On the other hand each V_j is isomorphic to some L_{k_j}, a (k_j+1)-dimensional irrep of highest weight k_j and highest weight vector $v_0^{(j)}$. On each $L_{k_j} = \bigoplus_{i=0}^{k_j} \mathbb{C} v_i^{(j)}$, e, f, h act as usual:

$$e \cdot v_i^{(j)} = v_{i-1}^{(j)}, \; i \geq 1 \qquad f \cdot v_i^{(j)} = (i+1)(k-i)\, v_{i+1}^{(j)}, \; i < k$$

$$e \cdot v_0^{(j)} = 0 \qquad f \cdot v_k^{(j)} = 0$$

$$h \cdot v_i^{(j)} = (k_j - 2i) \cdot v_i^{(j)}.$$

Now, we have $e^{k_j+1} = 0$ on each L_{k_j}, so $(\mathrm{ad}_X)^{k_j+1} = 0$ on V_j. Take $K = \max_j \{k_j\}$. Then $(\mathrm{ad}_X)^{K+1} = 0$ and thus ad_X is nilpotent as an endomorphism on $\mathfrak{g}$. □

From this same construction, we obtain that Y is a nilpotent and H is a semisimple element of $\mathfrak{g}$.

There is a stronger statement that works for any finite dimensional representation of the semisimple Lie algebra $\mathfrak{g}$.

Theorem 6: Let $\mathfrak{g}$ be a semisimple Lie algebra. If $X \in \mathfrak{g}$, then TFAE:

a) X is a semisimple (resp. nilpotent) element of $\mathfrak{g}$.

b) For every finite-dimensional representation (ρ, V) of $\mathfrak{g}$, $\rho(X)$ is a semisimple (resp. nilpotent) element of $\mathrm{End}(V)$.

c) If $\mathfrak{g}$ is a Lie subalgebra of $\mathrm{End}(V)$, then X is semisimple (resp. nilpotent) as an element of $\mathrm{End}(V)$.

Our goal is to prove a converse statement to Theorem 5.

Theorem 7 (Jacobson - Morozov)

Let $\mathfrak{g}$ be a semisimple Lie algebra and $X \in \mathfrak{g}$ be a nilpotent element. Then there exist $Y, H \in \mathfrak{g}$ such that $\{X, Y, H\}$ is a standard triple and X is nilpositive, Y is nilnegative and H is semisimple.

The proof for a more general semisimple Lie algebra will require a lot of work. We start with the case $\mathfrak{g} = \mathfrak{sl}_n$.

Example: $\mathfrak{g} = \mathfrak{sl}_3$, $X = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$.

Note that X is already a nilpotent matrix, written in its Jordan form. We need to find H such that $H \in \mathfrak{sl}_3$ so that ad_H acts semisimply on $\mathfrak{sl}_3$ and $[H, X] = 2X$. Writing

$$H = \begin{pmatrix} h_1 & 0 & 0 \\ 0 & h_2 & 0 \\ 0 & 0 & h_3 \end{pmatrix}, \text{ we have}$$

$$[H, X] = \begin{pmatrix} 0 & h_1 - h_2 & 0 \\ 0 & 0 & h_2 - h_3 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 2 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$$

and together with $h_1 + h_2 + h_3 = 0$, we obtain $h_1 = 2$, $h_2 = 0$, $h_3 = -2$.

Now, we need to find $Y \in \mathfrak{sl}_3$ such that $[X, Y] = H$. Note that Y is nilpotent and X is upper triangular. A good guess for Y is

$$Y = \begin{pmatrix} 0 & 0 & 0 \\ y_1 & 0 & 0 \\ 0 & y_2 & 0 \end{pmatrix}.$$

$$[X, Y] = \begin{pmatrix} y_1 & 0 & 0 \\ 0 & y_2 - y_1 & 0 \\ 0 & 0 & -y_2 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -2 \end{pmatrix}$$

so $y_1 = y_2 = 2$ and $Y = \begin{pmatrix} 0 & 0 & 0 \\ 2 & 0 & 0 \\ 0 & 2 & 0 \end{pmatrix}$.

Hence, $\{X, Y, H\}$ is the required standard triple.

Theorem 6: Let $X \in \mathfrak{sl}_n$ be a nilpotent element. Then there exist $Y, H \in \mathfrak{sl}_n$ such that $\{X, Y, H\}$ is a standard triple where X is nilpositive.

Proof: We know X is a nilpotent traceless matrix, so all its eigenvalues are zero. Because nilpotency is preserved in conjugacy classes, it is enough to prove the statement for X in its Jordan canonical form. As such, X can be written as

$$X = \begin{pmatrix} J_1 & & 0 \\ & J_2 & \\ 0 & & \ddots \; J_m \end{pmatrix},$$

where J_k are nilpotent Jordan blocks, with zeros in the diagonal and ones in the subdiagonal immediately above.

$$J_k = \begin{pmatrix} 0 & 1 & & & \\ & 0 & 1 & & \\ & & 0 & 1 & \\ & & & \ddots & \\ & & & & 0 \end{pmatrix}.$$

Under the basis $u_j = e_{j+1}$, $0 \leq j \leq k-1$, we have $X u_j = u_{j-1}$ and X acts like a raising operator. Since the subalgebra spanned by X, Y, H is isomorphic to $\mathfrak{sl}_2$, we can use what we know from the representation theory of $\mathfrak{sl}_2$.

We now need to find $H, Y \in \mathfrak{sl}_n$ with the desired properties.

i) Construction of H: We need $[H, X] = 2X$. Moreover, H should act diagonally on the basis vectors u_j:

$$H \cdot u_j = \lambda_j u_j.$$

Hence $[H, X]u_j = HXu_j - XHu_j = Hu_{j-1} - X\lambda_j u_j$
$= (\lambda_{j-1} - \lambda_j)u_{j-1}$

and $2Xu_j = 2u_{j-1}$,

so $\lambda_{j-1} - \lambda_j = 2 \Rightarrow \lambda_j = -2j + c$

Since $tr(H) = 0$, $\sum_{j=0}^{k-1} \lambda_j = 0 \Rightarrow -(k-1)k + kc = 0 \Rightarrow c = k-1$

And thus, $Hu_j = ((k-1) - 2j)u_j$. Note that this conclusion aligns perfectly with what we had before.

2) Construction of Y: We need Y to be a lowering operator on the u_j satisfying $[X, Y] = H$.

Let $Yu_j = y_{j+1}u_{j+1}$ and $Yu_{k-1} = 0$

Using the commutator above,

$$XY \cdot u_j - YX \cdot u_j = H \cdot u_j$$
$$X(y_{j+1}u_{j+1}) - Y(u_{j-1}) = ((k-1) - 2j)u_j$$
$$(y_{j+1} - y_j)u_j = ((k-1) - 2j)u_j$$

solving the recurrence $y_{j+1} - y_j = (k-1) - 2j$ with $y_k := 0$ produces $y_j = j(k-j)$, $0 \leq j \leq k-1$.

Define $Yu_j = j(k-j)u_{j+1}$. We need to check compatibility with $[H, Y] = -2Y$.

$$[H, Y]u_j = HYu_j - YHu_j$$
$$= H(j(k-j))u_{j+1} - Y((k-1)-2j)u_j$$
$$= [((k-1) - 2(j+1))(j(k-j)) - (((k-1)-2j)(j(k-j))u_{j+1}$$
$$= -2j(k-j)u_{j+1} = -2Yu_j,$$

as expected. □

Therefore, $\{X, Y, H\}$ is a standard triple in Sl_n.

Before we prove Theorem 7 for an arbitrary semisimple Lie algebra $\mathfrak{g}$, we state the following Lemma.

Consider the map $ad_X : \mathfrak{g} \to End(\mathfrak{g})$ and denote by $[X, \mathfrak{g}] = Im(ad_X)$ and $\mathfrak{g}^X = \{Z \in \mathfrak{g} : [X,Z] = 0\}$, the centralizer of X.

Lemma: For all $X \in \mathfrak{g}$, $[X, \mathfrak{g}] = (\mathfrak{g}^X)^\perp$, where orthogonality is taken with respect to the killing form.

Proof: Take $A \in [X, \mathfrak{g}]$. Then $\exists B \in \mathfrak{g}$ such that $[X, B] = A$. If $Z \in \mathfrak{g}^X$, then $[X, Z] = 0$. Since $\mathfrak{g}$ is semisimple, κ is nondegenerate.

$$\kappa(A, Z) = \kappa([X,B], Z) = -\kappa([B,X], Z) = -\kappa(B, [X,Z]) = -\kappa(B, 0) = 0.$$

Thus $[X, \mathfrak{g}] \subseteq (\mathfrak{g}^X)^\perp$.

Equality follows from checking dimensions:

κ is nondegenerate $\Rightarrow \dim \mathfrak{g}^X + \dim (\mathfrak{g}^X)^\perp = \dim \mathfrak{g} \Rightarrow \dim (\mathfrak{g}^X)^\perp = \dim \mathfrak{g} - \dim \mathfrak{g}^X$.

$\dim [X, \mathfrak{g}] + \dim \underbrace{\ker ad_X}_{\mathfrak{g}^X} = \dim \mathfrak{g} \Rightarrow \dim [X, \mathfrak{g}] = \dim \mathfrak{g} - \dim \mathfrak{g}^X$.

$$\therefore \quad [X, \mathfrak{g}] = (\mathfrak{g}^X)^\perp.$$

Proof of Theorem 7: We argue by induction on $\dim \mathfrak{g}$. If $\dim \mathfrak{g} = 3$, then $\mathfrak{g}$ must be isomorphic to $\mathfrak{sl}_2$, and the result follows. Suppose $\dim(\mathfrak{g}) > 3$. If X lies in a proper semisimple subalgebra $\mathfrak{a}$ of $\mathfrak{g}$, then by induction we can already find a standard triple $\{H, X, Y\}$ in $\mathfrak{a}$ with X as its nilpositive element. Thus, assume for the rest of the proof that X does not lie in a proper semisimple

subalgebra of $\mathfrak{g}$.

Claim 1: There exists $H' \in \mathfrak{g}$ such that $[H', X] = 2X$.

We start by showing $X \in (\mathfrak{g}^X)^\perp = [X, \mathfrak{g}]$. Indeed, if $Z \in \mathfrak{g}^X$, then

$$[\mathrm{ad}_X, \mathrm{ad}_Z] = \mathrm{ad}_{[X,Z]} = 0,$$

so ad_X and ad_Z commute.

X is nilpotent by assumption, so choose $r > 0$ such that $(\mathrm{ad}_X)^r = 0$. Since $[\mathrm{ad}_X, \mathrm{ad}_Z] = 0$,

$$(\mathrm{ad}_X \mathrm{ad}_Z)^r = (\mathrm{ad}_X)^r (\mathrm{ad}_Z)^r = 0, \text{ and}$$

$\mathrm{ad}_X \mathrm{ad}_Z$ is nilpotent. Any nilpotent endomorphism has trace zero, so

$$\kappa(X, Z) = \mathrm{Tr}(\mathrm{ad}_X \mathrm{ad}_Z) = 0 \Rightarrow X \in (\mathfrak{g}^X)^\perp = [X, \mathfrak{g}].$$

Take $M \in \mathfrak{g}$ such that $X = [X, M]$ and define $H' = -2M$. Then $[H', X] = -2MX + 2XM = 2[X, M] = 2X$.

Claim 2: There exists a semisimple element $H \in \mathfrak{g}$ such that $[H, X] = 2X$.

Let $H' = H'_s + H'_n$ be the Jordan decomposition of H'. We know $[H', X] = 2X$, or equivalently, $\mathrm{ad}_{H'}(X) = 2X$. Now $\mathrm{ad}_{H'} = \mathrm{ad}_{H'_s} + \mathrm{ad}_{H'_n}$, by Theorem 1, $\mathrm{ad}_{H'_s}$ and $\mathrm{ad}_{H'_n}$ are polynomials in ad_H with zero constant term and they stabilize the subspace $\mathbb{C}X$ (because $\mathrm{ad}_{H'}(X) = 2X \in \mathbb{C}X$). Finally, $\mathrm{ad}_{H'_n}$ acts on X by multiplying by zero. This happens because $\mathrm{ad}_{H'_n}$ is nilpotent, with some $r > 0$ satisfying

$(ad_{H_n'})^r = 0$. Then $ad_{H_n'} \cdot X = cX \Rightarrow [H_n', X] = cX$

$(ad_{H_n'})^r \cdot X = c^r \cdot X = 0 \Rightarrow c^r = 0 \Rightarrow c = 0$.

Hence $[H_n', X] = 0$, $[H_s', X] = 2X$, so we take $H = H_s'$.

Claim 3: H described in claim 2 is such that $H \in [X, g]$.

Suppose for now claim 3 is true and let's finish the proof of the theorem.

Let $Y' \in g$ be such that $[X, Y'] = H$.

Now, ad_H acts semisimply on g. We can write a decomposition of g into ad_H-eigenspaces:

$$g = \bigoplus_{i=1}^{k} g_{\lambda_i},$$

where $g_\lambda = \{Z \in g : [H, Z] = \lambda Z\}$. Write $Y' = Y_1' + \cdots + Y_k'$, with $Y_i' \in g_{\lambda_i}$.

Take any $Z \in g_{\lambda_i}$. Then $[H, Z] = \lambda_i Z$ and the Jacobi identity implies

$$\begin{aligned}[H, [X, Z]] &= [[H, X], Z] + [X, [H, Z]] \\ &= [2X, Z] + [X, \lambda_i Z] \\ &= (2 + \lambda_i)[X, Z].\end{aligned}$$

Hence $[X, Z] \in g_{\lambda_i + 2}$. That is, $[X, g_{\lambda_i}] \subseteq g_{\lambda_i + 2}$.

We know $ad_H(H) = 0$, so $H \in g_0$. Moreover,

$$H = [X, Y'] = \sum_{i=1}^{k} [X, Y_i'] \in g_0.$$

That implies all terms where $\lambda_i + 2 \neq 0$ must vanish. From this, $H = [H, Y_{-2}'] = -2Y_{-2}'$. Take $Y = Y_{-2}'$. and $\{H, X, Y\}$ is the desired standard triple.

Proof of Claim 3: We argue by contradiction.
Suppose $H \notin [X, \mathfrak{g}]$, or equivalently, $H \notin (\mathfrak{g}^X)^\perp$.
Hence, there exists $Z \in \mathfrak{g}^X$ s.t. $\kappa(H, Z) \neq 0$

On the other hand, $\forall Z \in \mathfrak{g}^X$, we have $ad_H \cdot Z \in \mathfrak{g}^X$.
Indeed,

$$[X, ad_H \cdot Z] = [X, [H, Z]] =$$
$$= [[Z, X], H] + [[X, H], Z]$$
$$= 0 + [-2X, Z] = 0 - 2[X, Z] = 0,$$

so ad_H leaves $\mathfrak{g}^X$ invariant. Further, we know ad_H must act semisimply, so we can decompose $\mathfrak{g}^X$ into ad_H eigenspaces: $\mathfrak{g}^X = \bigoplus_{i=1}^{t} \mathfrak{g}^X_{\tau_i}$, $\tau_i \in \mathbb{C}$.

Note that $\mathfrak{g}^X_0 = \{ Z \in \mathfrak{g}^X \mid [H, Z] = 0 \} = (\mathfrak{g}^X)^H$, so

$$\mathfrak{g}^X = (\mathfrak{g}^X)^H \oplus \bigoplus_{\tau_i \neq 0} \mathfrak{g}^X_{\tau_i}.$$

Also, invariance of the killing form implies

$$\kappa(H, [H, \mathfrak{g}^X]) = \kappa([H, H], \mathfrak{g}^X) = 0.$$

If $Z \in \mathfrak{g}^X_{\tau_i}$ with $\tau_i \neq 0$, then

$$0 = \kappa(H, [H, Z]) = \kappa(H, \tau_i Z) = \tau_i \kappa(H, Z)$$
$$\Rightarrow \kappa(H, Z) = 0 \Rightarrow Z \in (\mathfrak{g}^X_{\tau_i})^\perp, \ \forall \tau_i \neq 0.$$

Since we want $Z \in \mathfrak{g}^X$ such that $\kappa(H, Z) \neq 0$, we conclude $\exists Z \in (\mathfrak{g}^X)^H$ such that $\kappa(H, Z) \neq 0$.

If Z is nilpotent, a similar argument to that used in the proof of Claim 1 shows $ad_H ad_Z$ is nilpotent and $\kappa(H, Z) = \mathrm{Tr}(ad_H ad_Z) = 0$.

Hence $Z_s \neq 0$, and by the Jordan Decomposition Theorem, Z_s is a nonzero semisimple element in $(g^X)^H$.

Then g^{Z_s} is reductive (This follows from the fact that if $M \neq 0$ is semisimple, g^M is reductive. Can be proven by root space decomposition), so $[g^{Z_s}, g^{Z_s}]$ is a semisimple subalgebra of g. It is a proper subalgebra, since $g^{Z_s} = g$ only if $Z_s = 0$.

Now, $Z_s \in (g^X)^H$, $[H, Z_s] = 0 \Rightarrow H \in g^{Z_s}$.

$Z_s \in g^X$, $[X, Z_s] = 0 \Rightarrow X \in g^{Z_s}$.

Hence $2X = [H, X] \in [g^{Z_s}, g^{Z_s}]$, so the nilpotent element X is in a proper semisimple subalgebra of g, contradicting our initial assumption. Therefore, $H \in [X, g]$ and the proof is finished.

Adjoint Orbits

One of our goals is to classify the semisimple and nilpotent elements in any semisimple Lie algebra g.

To each Lie algebra g we can associate its adjoint group G_{ad}, defined as the connected subgroup of $GL(g)$ with Lie algebra ad_g.

In the case where g is semisimple, this group can be realized more explicitly. We start with

$$\mathrm{Aut}(g) = \{\phi \in GL(g) : [\phi(X), \phi(Y)] = \phi([X,Y]),\ \forall X, Y \in g\}.$$

Note that for $\phi \in \mathrm{Aut}(g)$ and $X \in g$ we have

$$\phi \cdot ad_X \cdot \phi^{-1} = ad_{\phi(X)},$$

Indeed, for all $Y \in \mathfrak{g}$,

$$(\phi \cdot \mathrm{ad}_X \cdot \phi^{-1}) \cdot Y = \phi \cdot ([X, \phi^{-1}(Y)]) = [\phi(X), Y] = \mathrm{ad}_{\phi(X)} \cdot Y.$$

This implies nilpotence or semisimplicity of X are equivalent to nilpotence or semisimplicity of $\phi(X)$, $\phi \in \mathrm{Aut}(\mathfrak{g})$.

Now let $\mathfrak{g}$ be a semisimple Lie algebra and let G be any connected semisimple Lie group with Lie algebra $\mathfrak{g}$.

For each $A \in G$, we define the map $\mathrm{Ad}_A : \mathfrak{g} \to GL(\mathfrak{g})$ as $\mathrm{Ad}_A \cdot X = AXA^{-1}$. It is straightforward to check that $\mathrm{Ad}_A([X,Y]) = [\mathrm{Ad}_A \cdot X, \mathrm{Ad}_A \cdot Y]$, i.e., $\mathrm{Ad}_A \in \mathrm{Aut}(\mathfrak{g})$. From this, we define the adjoint representation of G as $\mathrm{Ad} : G \to \mathrm{Aut}(\mathfrak{g})$, $A \mapsto \mathrm{Ad}_A$.

With this at hand, we can define G_{ad} to be the $G_{ad} = \mathrm{Ad}_G = \mathrm{Image}(\mathrm{Ad})$. Another definition is $G_{ad} = \mathrm{Aut}(\mathfrak{g})^0$, where $\mathrm{Aut}(\mathfrak{g})^0$ is the identity component of $\mathrm{Aut}(\mathfrak{g})$. These two definitions are compatible by the following Lemma.

Lemma: Let G be a connected semisimple complex Lie group with Lie algebra $\mathfrak{g}$. Then $G_{ad} = \mathrm{Ad}_G = \mathrm{Aut}(\mathfrak{g})^0$.

Proof: Ad and ad are related by

$$\exp(\mathrm{ad}_X) = \mathrm{Ad}_{\exp(X)}.$$

Start with $G_{ad} = \mathrm{Ad}_G = \mathrm{Image}(\mathrm{Ad})$ and pick $X \in \mathfrak{g}$. Then by definition of a Lie algebra, we know $\exp(tX) \in G$ for all $t \in \mathbb{R}$. Thus, $\mathrm{Ad}_{\exp(tX)} \in \mathrm{Ad}_G$, or equivalently, $\exp(t\,\mathrm{ad}_X) \in \mathrm{Ad}_G$ for all $t \in \mathbb{R}$. Again by definition of a Lie algebra, it follows that $\mathrm{ad}_X \in \mathrm{Lie}(\mathrm{Ad}_G)$. This proves $\mathrm{ad}_{\mathfrak{g}} \subseteq \mathrm{Lie}(\mathrm{Ad}_G)$.

Conversely, take $Z \in \text{Lie}(\text{Ad}_G)$. Then $\exp(tZ) \in \text{Ad}_G$ for all $t \in \mathbb{R}$. Take $A(t) \in G$ with $A(0) = I$ and write $\exp(tZ) = \text{Ad}_{A(t)}$. Taking derivatives at $t = 0$ and denoting $A'(0) = X$, we have, for all $Y \in \mathfrak{g}$,

$$\frac{d}{dt}\left(\exp(tZ)\right)\Big|_{t=0} \cdot Y = \frac{d}{dt}\left(\text{Ad}_{A(t)} \cdot Y\right)\Big|_{t=0}$$

$$Z \cdot Y = \frac{d}{dt}\left(A(t) Y A(t)^{-1}\right)\Big|_{t=0}$$

$$Z \cdot Y = XY - YX = \text{ad}_X \cdot Y,$$

so $Z = \text{ad}_X$. We also have $X = A'(0) \in \mathfrak{g}$ as it is the derivative of a smooth curve at $t = 0$. So $Z \in \text{ad}_{\mathfrak{g}}$ and thus $\text{Lie}(\text{Ad}_G) \subseteq \text{ad}_{\mathfrak{g}}$.

This proves the earlier statement that $\text{Lie}(\text{Ad}_G) = \text{ad}_{\mathfrak{g}}$.

It is a standard fact that $\text{Lie}(\text{Aut}(\mathfrak{g})) = \text{Der}\,\mathfrak{g}$ and that for semisimple $\mathfrak{g}$, $\text{ad}_{\mathfrak{g}} = \text{Der}\,\mathfrak{g}$. Since $\text{Aut}(\mathfrak{g})^0$ and $\text{Aut}(\mathfrak{g})$ have the same Lie algebra, it follows that $\text{Aut}(\mathfrak{g})^0$ and Ad_G are two connected subgroups of $GL(\mathfrak{g})$ with the same Lie algebra $\text{ad}_{\mathfrak{g}}$. Such subgroup is unique, so $\text{Aut}(\mathfrak{g})^0 = \text{Ad}_G = G_{ad}$.

□

Another important consequence of the Lemma is that the center of G satisfies $Z(G) = \ker(\text{Ad})$.

Ex.: $\mathfrak{g} = \mathfrak{sl}_n$ has Lie group $G = SL_n$. Using the adjoint representation $\text{Ad} : SL_n \longrightarrow \text{Aut}(\mathfrak{sl}_n)$, we have $\ker(\text{Ad}) = Z(SL_n) = \{\pm I\}$, so that $G_{ad} \cong SL_n / \{\pm I\} = PSL_n$.

Given $\mathfrak{g}$ a semisimple Lie algebra and $X \in \mathfrak{g}$ we consider the action of the adjoint group on the vector space $\mathfrak{g}$ and define the adjoint orbit through X by

$$\mathcal{O}_X = G_{ad} \cdot X = \{\phi(X) \mid \phi \in G_{ad}\}.$$

An adjoint orbit is called nilpotent (resp. semisimple) if X is nilpotent (resp. semisimple). Since $\phi(X)$ is nilpotent iff X is nilpotent, it follows that an orbit is nilpotent iff each element is nilpotent (the same for semisimple). Therefore, to classify the nilpotent and semisimple element in $\mathfrak{g}$, it is enough to classify their orbits.

Let $\{X, Y, H\}$ be a nonzero standard triple in the semisimple Lie algebra $\mathfrak{g}$. Note that the set of all nonzero standard triples in $\mathfrak{g}$ is invariant under the action of G_{ad}. Denote by A_{triple} the set of all G_{ad} - conjugacy classes of nonzero standard triples in $\mathfrak{g}$ and denote by S_{dist} the set of (semisimple) orbits of the form $\mathcal{O}_H$, where H is the semisimple element of a standard triple. From now on we call the set S_{dist} the set of distinguished semisimple orbits in $\mathfrak{g}$.

In the process of classifying nilpotent orbits in a semisimple Lie algebra we go through three steps.

1) There is a one-to-one correspondence between the sets A_{triple} and the set of nonzero nilpotent orbits in $\mathfrak{g}$.

2) There is a one-to-one correspondence between the sets A_{triple} and S_{dist}.

3) There is a one-to-one correspondence between the set of nilpotent orbits in $\mathfrak{g}$ and certain weighted Dynkin diagrams of distinguished semisimple elements. In particular, there are at most $3^{\text{rank}(\mathfrak{g})}$ such diagrams, and thus, at most the same number of nilpotent orbits in $\mathfrak{g}$.

We will focus our work on classifying all the nilpotent orbits in sl_n and their connection to these special weighted Dynkin diagrams. In the case of sl_n, these diagrams are of type A_{n-1}.

Classification of Nilpotent Orbits in $\mathfrak{g} = sl_n$.

Given $\lambda \vdash n$, $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_k)$, $\lambda_i \neq 0$, we can form the block diagonal matrix X_λ

$$X_\lambda = \begin{pmatrix} J_{\lambda_1} & & & 0 \\ & J_{\lambda_2} & & \\ & & \ddots & \\ 0 & & & J_{\lambda_k} \end{pmatrix},$$

where J_{λ_i} are elementary Jordan blocks of size $\lambda_i \times \lambda_i$. Each X_λ is a nilpotent element of sl_n. We look at the orbits

$$\mathcal{O}_\lambda = \mathcal{O}_{X_\lambda} = PSL_n \cdot X_\lambda.$$

For any nilpotent element in sl_n, its Jordan blocks can be rearranged in decreasing order of size with the aid of similarity transformations.

Note that if $\lambda \neq \mu$ are two distinct partitions of n, then $\mathcal{O}_\lambda$ and $\mathcal{O}_\mu$ are disjoint by the uniqueness of the Jordan form. Further, by Linear algebra, each X has a Jordan canonical form X_λ for some $\lambda \vdash n$. It is not difficult to see that if X_λ is conjugate to X in GL_n, then they are also PSL_n-conjugates. This shows there exists a one-to-one correspondence between nilpotent orbits in sl_n and the set of partitions of n, and in particular, the set of nilpotent orbits is finite. Note that the zero orbit corresponds to (1^n).

Weighted Dynkin Diagrams of Type A_{n-1}

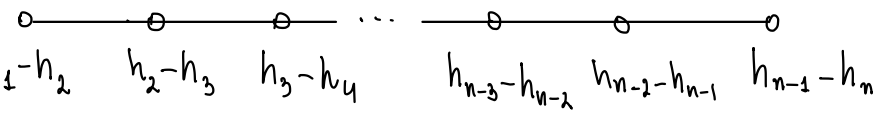
Let $\lambda \vdash n$, with $\lambda = (\lambda_1, \ldots, \lambda_k)$ and $\lambda_k > 0$. We learned that there exists a nilpotent orbit $\mathcal{O}_\lambda$ of $\mathfrak{sl}_n$ labeled by λ. The proof of the Jacobson-Morozov theorem for $\mathfrak{sl}_n$ produced a recipe on how to construct a standard triple associated with $\mathcal{O}_\lambda$. Denote such triple by $\{X_\lambda, Y_\lambda, H_\lambda\}$, where X_λ is a direct sum of nilpotent Jordan blocks of sizes $\lambda_1, \ldots, \lambda_k$ and H_λ is semisimple and a direct sum of diagonal matrices of the following form.

$$H_\lambda = \begin{pmatrix} D_{\lambda_1} & & 0 \\ & D_{\lambda_2} & \\ 0 & & \ddots \\ & & & D_{\lambda_k} \end{pmatrix}, \quad \text{where} \quad D_{\lambda_i} = \begin{pmatrix} \lambda_i - 1 & & & \\ & \lambda_i - 3 & & \\ & & \ddots & \\ & & & -\lambda_i + 1 \end{pmatrix}$$

The diagonal matrices of trace zero form a Cartan subalgebra of $\mathfrak{sl}_n$. The Weyl group is S_n and it acts on diagonal matrices by permuting the diagonal entries. To construct the corresponding diagram, we take a representative in the equivalence class of H_λ, called the dominant element and denoted by $\tilde{H}_\lambda$, for which the diagonal entries are organized in decreasing order. That is,

$$\tilde{H}_\lambda = \begin{pmatrix} h_1 & & & \\ & h_2 & & \\ & & \ddots & \\ & & & h_n \end{pmatrix}, \quad \text{with} \quad h_1 \geq h_2 \geq \cdots \geq h_n.$$

The weighted Dynkin diagram associated with $\mathcal{O}_\lambda$, denoted by $\Delta(\mathcal{O}_\lambda)$, is given by

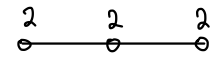
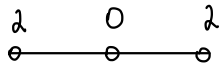
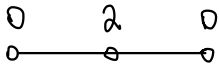
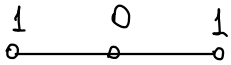
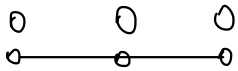


Since the simple roots for A_{n-1} are $\alpha_i = \varepsilon_i - \varepsilon_{i+1}$, the labels are $a_i = \alpha_i(H_\lambda) = h_i - h_{i+1}$, $1 \leq i \leq n-1$.

Note that the weights of $\Delta(\mathcal{O}_\lambda)$ are symmetric about the center of the diagram.

The weights in the Dynkin diagrams of $\mathfrak{sl}_n$ (and in a general semisimple Lie algebra $\mathfrak{g}$) are the elements of the set $\{0,1,2\}$. This justifies the $3^{\text{rank}(\mathfrak{g})}$ upper bound mentioned before.

Example: Take $n=4$, i.e., $\mathfrak{g} = \mathfrak{sl}_4$. Nilpotent orbits are labeled by partitions of 4.

$\mathcal{O}_\lambda$	H_λ	$\tilde{H}_\lambda$	$\Delta(\mathcal{O}_\lambda)$
$\mathcal{O}_{(4)}$	$\text{diag}(3,1,-1,-3)$	$\text{diag}(3,1,-1,-3)$	
$\mathcal{O}_{(3,1)}$	$\text{diag}(2,0,-2,0)$	$\text{diag}(2,0,0,-2)$	
$\mathcal{O}_{(2,2)}$	$\text{diag}(1,-1,1,-1)$	$\text{diag}(1,1,-1,-1)$	
$\mathcal{O}_{(2,1,1)}$	$\text{diag}(1,-1,0,0)$	$\text{diag}(1,0,0,-1)$	
$\mathcal{O}_{(1,1,1,1)}$	$\text{diag}(0,0,0,0)$	$\text{diag}(0,0,0,0)$	

Now, there is a one-to-one correspondence between weighted Dynkin diagrams and nilpotent orbits of $\mathfrak{sl}_n$. Given the weights $(w_1, w_2, w_3, \ldots, w_{n-1})$, we start by picking the largest entry $l_1 = \max_i(w_i)$. We then subtract 2 successively until we obtain the set $\{l_1, l_1-2, \ldots, -l_1+2, -l_1\}$, whose cardinality is λ_1. Next, we take the second largest weight $l_2 = \max_i\{w_i\} \setminus \{l_1, l_1-2, \ldots, -l_1+2, -l_1\}$ and obtain the set $\{l_2, l_2-2, \ldots, -l_2+2, -l_2\}$, whose cardinality is λ_2, and so on. In that way, we recover the partition $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_k)$ and the nilpotent orbit $\mathcal{O}_\lambda$.